

USING BOVINE MEAT INSPECTION DATA FOR SYNDROMIC SURVEILLANCE: INNOVATIVE STATISTICAL APPROACH FOR DEFINING SYNDROMES

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SUMMARY

Defining syndromes when performing syndromic surveillance is challenging. Definitions are commonly based on expert opinion consensus, not always easy to reach. Syndrome definition based on meat inspection data is particularly challenging due to the type of data collected regarding zootechnical information (breed, age, sex) and condemnation (portions and motives). Furthermore, each animal can have more than one condemnation motive. This study presents a statistical approach for syndrome definition based on both Multiple Factorial Analysis and clustering methods to define groups of condemnation motives, portions and zootechnical characteristics. Data from ten French bovine slaughterhouses were used, involving 1,937,917 cattle. Twelve clusters considered as stable according to year of slaughter and slaughterhouse were identified that deal with animal or public health surveillance, but also animal welfare, quality of the slaughtering process, management issues and evaluation of economical impact on farmers.

INTRODUCTION

Slaughterhouses, a focal point of farm animals, are a potential source of data not otherwise available. Both zootechnical data (age, breed, sex) and condemnation data (condemnation motives and portions) can be collected. At this time, meat inspection data are under-used for purposes other than food safety. The recent development of slaughterhouse databases in several countries makes these data accessible for cattle health monitoring (Alton et al., 2010; Ruoho et al., 2010; Dupuy et al., 2012b).

However, as the main objective of meat inspection is food safety and not animal health surveillance, the aetiological diagnosis is often not reached. Meat inspection data have a variable specificity from non specific lesions (e.g. abscess) to very specific ones (e.g. liver fluke).

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Meat inspection and epidemiological surveillance

According to the type of meat inspection data used, different types of epidemiological surveillance could be performed (Table 1). Targeted surveillance (i.e. targeted objective) is focused on the surveillance of a pre-defined disease or group of diseases whereas non-targeted surveillance (i.e. non-targeted objective) aims at detecting unknown or emergent diseases. Syndromic surveillance can be either targeted or non-targeted according to the nature of the indicator monitored.

Table 1. Types of epidemiological surveillance according to the objective and examples of syndrome definition based on meat inspection data

Targeted objective	Non-targeted objective
Syndromic surveillance <i>e.g. lung lesions for respiratory diseases</i>	Syndromic surveillance <i>e.g. proportion of condemned cattle</i>
Traditional surveillance <i>e.g. muscular cysticercosis, fascioliasis</i>	

Impact of the type of epidemiological surveillance on syndrome definition method

The methodology used to define a “case” depends on the type of surveillance. For traditional surveillance, the case definition method is based on existing referential or expert definition, then relevant data are collected to meet the objective. For syndromic surveillance, available data are usually collected for another purpose and the epidemiologist usually has no power to modify the way data are collected, so using these data can be challenging. No referential of syndrome definition based on meat inspection data exists. Referring to complexity of the data (i.e. large diversity of motives and locations of condemnations), the definition through an expert process is difficult. The objective of this study was to implement a statistical approach to define groups of animals according to their zootechnical characteristics and lesions observed during the slaughtering process that could be used to define relevant syndromes to monitor.

MATERIALS AND METHODS

Data from ten French bovine slaughterhouses collected in real-time during the slaughtering process were available through the *Nergal-Abattoir* project implemented by the French ministry of Agriculture. Among the 1,937,917 cattle slaughtered in these slaughterhouses from July 2005 to December 2010, 381,186 (19.7%) had at least one part of the carcass condemned and were included in this study. For each bovine, the database contained: identification number, dates of birth and slaughter, last farm location, sex, breed, signs observed during ante-mortem inspection, condemnation motives and locations. All information registered during the slaughtering process and used to create variables for analyses are presented in Table 2. The ages of cattle were classified into six categories: 0-8 months, 8-24 months old, 2-3.5 years old, 3.5-5 years old, 5-10 years old and ≥ 10 years old. Some modalities of variables were grouped to avoid low numbers in categories that could

create instability in data analyses. The 264 different condemnation motives of the *Nergal-Abattoir* system were merged into 57 categories of condemnation motives according to their biological similarities to be able to compare data among slaughterhouses when the levels of detail of the motive were different (e.g. abscess, multiple abscess, local abscess were merged into abscess). The condemnation portions were merged into 12 modalities (whole carcass, part of carcass, skin, head, udder, heart, lungs, liver, kidneys, viscera, tongue and thymus).

Table 2. Steps of the slaughtering process, data collected and variables created for analysis

Slaughtering step	Data collected	Variables
Identification check :	Animal ID/Farm ID	Farm location (Region)
Registration of zootechnical information	Sex (Male/Female)	Sex: Female/Castrated
	Castrated (Yes/no)	male/Non-castrated male
	Breed (67 modalities)	Production type (Dairy/Beef/Mixed)
	Date of birth, Date of slaughter	Age category, Year and Month of slaughter
Ante-mortem inspection (AMI)	Signs observed (42 modalities)	AMI abnormality : Yes/No
Post-mortem inspection (PMI)	Condemnation motives: one for each location (264 modalities)	Condemnation motives (57 modalities)
	Condemnation portion (916 modalities)	Condemnation portions (12 modalities)

Multiple Factor Analysis (MFA) associated with clustering methods was used to establish groups of condemned cattle. Calculations were performed with R software (R Development Core Team, 2010), “FactomineR” and “cluster” packages. The steps of data analysis are presented in Fig.1.

Active variables that contributed to the construction of factorial axes of MFA were both zootechnical characteristics (sex, age category, production type) of condemned cattle and condemnation data (motives and portions). Year of slaughter, month of slaughter, farm location, presence of clinical signs during ante-mortem inspection, abattoir identification number, condemnation motives and portions with a proportion lower than 1% were used as illustrative variables i.e. variables that did not contribute to the construction of factorial axes but helped interpreting results of MFA. Clustering of bovines was performed from their principal coordinates by computing the classical Euclidean distance between these coordinates (Bécue-Bertaut & Pagès, 2008). Considering the large amount of data a previous K-means was performed before Hierarchical Ascendant Clustering (HAC). The generalised Ward’s criterion was used. The partition was determined considering the hierarchical tree and according to the biological meaning of the clusters (Lebart et al., 2006). The HAC was consolidated by a K-means performed on the centers of the HAC clusters.

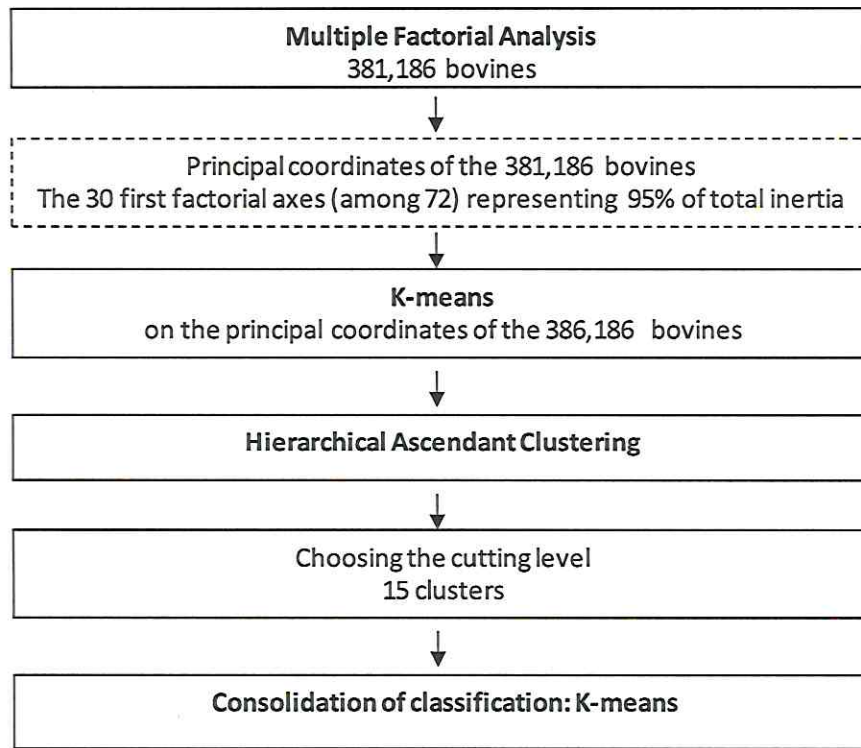


Fig.1 Steps of statistical data analysis

Stability of defined clusters was evaluated on 15 subsets of data according to year of slaughter (2006 to 2010: 5 subsets) and slaughterhouse (10 subsets) through the same process of analyses, i.e. MFA, K-means and HAC consolidated by K-means. Stability was defined as the proportion of subset analyses in which such a cluster was identified. Clusters with stability higher than 50% were considered as stable.

Description and interpretation of the clusters were based on both active and illustrative variables using V-Test values (Lebart et al., 2006). V-test of each modality of variables refers to the comparison between the proportion of cattle concerned by this modality in the cluster and the proportion of cattle with the same modality in the whole dataset formulated by a number of standard deviations of Gaussian law. Thus, the higher the difference between these two proportions is, the higher the absolute value of the V-test.

RESULTS

Descriptive analysis

According to the slaughterhouse, the mean number of cattle slaughtered each day varied from 122 to 543. The proportion of bovines having at least one portion of the carcass condemned varied from 9.6% to 35.9% according to slaughterhouses. Among the 381,186 cattle included in the study, (i.e. with condemnation), 70.0% were female, 25.5% non-castrated male and 4.5% castrated male. Beef cattle represented 44.3% of condemned cattle, dairy cattle 35.3% and mixed cattle 20.5%. The highest age category of condemned cattle was the 5- to 10-year-old age category (37.4%). The description of the 12 listed condemnation portions showed that liver condemnation concerned 67.5% of the cattle with at

least one portion of the carcass condemned. The condemnation of the whole carcass concerned 3.4% of condemned cattle. The most frequent condemnation motives were abscess (18.8%) and liver fluke (15.1%).

Multiple factor analysis and clustering

Results of the different steps of the data analysis suggested four possible partitions in 6, 9, 15 or 16 clusters based on the height of the HAC dendrogram. The 15 clusters partition was selected based on the biological significance of the clusters.

The clusters 1, 2, 4, 5, 7, 8, 9, 10, 12, 13, 14 and 15 gathered between 1.3% and 2.4% of all condemned cattle. Clusters 6 and 11 were bigger, with respectively 4.1% and 6.0% of the condemned cattle. Cluster 3 gathered almost two thirds (65.7%) of the condemned cattle. Among the 15 clusters, clusters 5, 6 and 10 were considered as not stable and are not presented in this paper. The main variables that characterised the other 12 clusters are presented in Table 3.

Table 3. Main characteristics of the 12 stable clusters

Cluster	Main characteristic variables	Cla/Mod ^a	Mod/Cla ^b	Global ^c	v.test
1 (n=9,268)	Fecal contamination	82.4	100	3.0	+∞
	Lungs	10.9	81.3	18.1	+∞
	Heart	14.8	61.5	10.1	+∞
2 (n=7,215)	Bronchopneumonia	15.4	28.7	3.5	+∞
	Pericarditis	87.1	100	2.2	+∞
	Lungs	4.8	46.0	18.1	+∞
	Heart	18.7	99.7	10.1	+∞
	Dairy cattle	2.8	51.3	35.3	28.2
3 (n=250,256)	Macular telangiectasia	87.7	11.1	8.3	+∞
	Liver fluke	89.0	25.6	18.9	+∞
	Sclerosis	81.0	12.5	10.1	+∞
	Abscess	74.5	26.7	23.6	+∞
	Liver	76.9	79.1	67.5	+∞
4 (n=8,736)	Peritonitis	63.5	100	3.6	+∞
	Abscess	4.2	43.0	23.6	+∞
	Part of the carcass	9.7	54.2	12.8	+∞
7 (n=5,120)	Local muscular cysticercosis	99.8	100	1.4	+∞
	Head	12.9	75.4	7.9	+∞
	Heart	4.4	32.8	10.1	+∞
	(2 -3.5 years old)	3.0	29.9	13.3	31.3
	Castrated male	4.7	15.5	4.5	30.5
8 (n=9,095)	Other deteriorations ^d	86.8	100	2.8	+∞
9 (n=8,993)	Steatosis	83.9	100	2.8	+∞
	Kidneys	4.6	37.8	19.2	+∞
	Liver	3.1	87.9	67.5	+∞
	Dairy cattle	4.0	60.2	35.3	+∞
	(5-10 years old)	3.3	52.5	37.4	29.5
11 (n=22,690)	Ante-mortem abnormality	27.8	18.3	3.9	+∞
	Arthritis	86.1	43.1	3.0	+∞
	Infiltration	83.9	49.0	3.5	+∞
	Part of the carcass	46.1	99.2	12.8	+∞
12 (n=7,383)	Myopathy	94.9	100	2.0	+∞
	Part of the carcass	15.1	99.7	12.8	+∞
	Ante-mortem abnormality	5.3	10.7	3.9	25.3

13 (n=8,778)	Bronchopneumonia	65.3	100	3.5	+∞
	Lungs	12.7	99.8	18.1	+∞
	(8- 24 months old)	5.9	52.0	20.3	+∞
	Non-castrated male	5.0	55.3	25.5	+∞
14 (n=7,944)	Emphysema	94.2	100	2.2	+∞
	Lungs	11.5	100	18.1	+∞
	Dairy cattle	4.2	71.4	35.3	+∞
	(5-10 years old)	3.1	55.1	37.4	32.3
15 (n=5,452)	Meat with abnormal maturation	98.1	100	1.5	+∞
	Ante-mortem abnormality	11.1	30.2	3.9	+∞
	Whole carcass	41.6	99.8	3.4	+∞

^a Proportion of cattle having this modality that were in the cluster e.g. 15.4% of cattle that had a lesion of bronchopneumonia in the whole population were in the cluster 2.

^b Proportion of cattle in the cluster having the modality e.g. 28.7% of the cattle in the cluster 2 had a lesion of bronchopneumonia

^c Proportion of cattle in the global population having the modality e.g. 3.5% of the whole population of cattle with condemnation had a lesion of bronchopneumonia.

^d Including haemorrhagic lesions of lungs or muscles linked to problems in the slaughtering process; superficial and deep putrefaction.

DISCUSSION

The objective of this study was to determine a typology of bovines with condemnation in order to define syndromes based on a statistical approach.

Clusters interpretation: typology of condemned cattle

The interpretation of each of the 12 clusters showed that meat inspection data can be used beyond food safety. The clusters describing various types of condemnation referred to five different fields of interpretation in relationship with farming and abattoir activities: animal health, animal welfare, breeding practice, public health and quality of the slaughtering process (Table 4). The three types of epidemiological surveillance in Table 1 could be addressed by monitoring these syndromes: traditional surveillance with cluster 7 (cysticercosis), targeted syndromic surveillance with for instance cluster 4 (chronic peritonitis lesions), non-targeted syndromic surveillance by monitoring the percentage of cattle in each cluster to detect changes in trends.

Limits of this typology

The typology of lesions presented in this study was based on French meat inspection data and was thus linked to the animal health and welfare situation in France. It could be interesting to build a similar typology based on meat inspection data from other countries to (i) highlight common clusters (i.e. groups of lesions) for which exchanges of information and knowledge could be relevant and (ii) try to understand why adjacent countries do not always share the same clusters.

As disease presence and patterns can change over years within a country, it could be useful to perform the description of groups of lesions based on the statistical method presented again when the health status of the country seems to have changed or after a defined period of time. This change would be detected by an important modification in trends of the proportion of cattle in each cluster.

Table 4. Interpretation and field of interest of stable statistical clusters identified

Cluster	Interpretation	Field
1	Fecal contamination of heart or lungs linked with failure during evisceration step	Quality of slaughtering process
8	Deteriorations linked to failure in the slaughtering process	
2	Pericarditis and bronchopneumonia which could be linked to traumatic reticulo-pericarditis	Breeding practice such as feeding
9	Fatty liver syndrome	
14	Lung emphysema : farmer's lung	
3	Chronic liver lesions	Breeding practice
4	Chronic peritonitis lesions which could be linked to previous infection	
11	Arthritis	Animal health, Animal welfare
12	Myopathy (muscle crush syndrome)	Animal welfare, Breeding practice
13	Bronchopneumonia	Animal health, Breeding practice
15	Dark Firm Dry meat	Animal welfare (transport management)
7	Cysticercosis	Public health (zoonotic disease)

Syndrome definition methods according to the type of syndromic surveillance

As stated in the introduction, the syndrome definition method is different according to the type of syndromic surveillance (Table 5). For targeted syndromic surveillance (i.e. the objective of the system is the detection of a known disease or group of diseases), three approaches could be used: (1) definition based on an existing referential which, to our knowledge, does not exist for meat inspection data; (2) definition based on expert opinion consensus; and (3) definition based on both expert opinion and statistical typology of existing groups of lesions performed on historical data. Indeed reaching a consensus among experts based on a wide list of lesions associated or not with zootechnical criteria is challenging. It seems relevant to start with a statistical analysis of groups of lesions and zootechnical characteristics as support for further discussion among experts.

For non-targeted syndromic surveillance (i.e. the objective of the system is the detection of unknown diseases), generic indicators need to be defined. Two approaches have been identified. First, based on the hypothesis that each disease has a dedicated organ tropism, more or less specific, and evolution stages of lesions, more or less acute, it seems relevant to assume that monitoring the proportion of condemned cattle according to the extent/location of condemnation (whole carcass, part of the carcass, heart, liver...), whatever the nature of the lesion, could allow the detection of an emerging disease. Second, based on the fact that each disease is characterised by group(s) of lesions and that it affects a susceptible bovine population characterised by zootechnical criteria, it seems relevant to assume that most cattle affected with an emerging disease will have similar lesions and zootechnical characteristics. Using the statistical syndrome definition, each condemned bovine added in the dataset used to define clusters could be attributed to a cluster by determining its MFA-derived

representation and the cluster it belongs to. It is thus probable that most of these cattle will be attributed to the same cluster, making its proportion abnormally increasing.

Table 5. Advantages and limits of the different syndrome definition methods

Type of syndromic surveillance	Syndrome definition method	Advantages	Limits
Targeted	Based on existing referential	Easy to use Easier for comparability among systems Cheap	Need of an existing referential Accessibility of the data as defined in the referential
	Based on expert opinion consensus	Classical and known approach	Difficulty to reach consensus Depends on the experts recruited Cost for expert group Accessibility of the data as defined by experts
	Statistical definition associated with expert biological interpretation	Based on both historical data and expert opinion Help for reaching consensus Could identify syndromes not identified by experts Start point is available data which guarantee accessibility to data needed	Historical data needed Depends on the experts recruited Cost for expert group
Non-targeted	Statistical definition	Based on historical data/the reality Possible for complex and large sets of data	Historical data needed
	Expert definition: Generic indicator	Easy to define	Not able to take into account the complexity of the data

Added value and limits of meat inspection data for bovine surveillance

Using meat inspection data for animal health surveillance is tempting. Indeed these data are not otherwise available which makes the detection of diseases not detectable by other types of surveillance (e.g. cysticercosis) possible. The slaughterhouse is the focal point of most cattle thus meat inspection data deals with nearly the entire bovine population. More and more countries are implementing databases to collect these data. This could allow the

implementation of a nearly exhaustive surveillance system. As demonstrated in this study, meat inspection data deals not only with food safety but also with animal health and welfare, breeding practice and the slaughtering process, making these data potentially useful for several objectives and stakeholders (veterinary services, farmers, etc.).

Limits of meat inspection data could however be identified. As explained before, these data are collected for food safety purpose which has an impact on the specificity of these data if used for another purpose. It is also difficult to make the link between the lesions observed at the slaughterhouse and the location of the onset of the disease (last farm or not?) especially for chronic lesions. It could create important biases in spatial analyses if it were not taken into account. As with all systems collecting data through different persons, there could be differences according to the person performing the meat inspection. Likewise, differences could exist according to the slaughterhouse in terms of sensitivity and specificity of the inspection. An EFSA European project called “Contribution of meat inspection to animal health surveillance” was performed to quantify the effectiveness of meat inspection to detect a list of diseases and welfare conditions (Dupuy et al., 2012a). The probability of detection of a disease (i.e. probability of detection of a sign during ante-mortem inspection or lesion during post-mortem inspection) depends on a lot of factors that need to be taken into account when using meat inspection data such as the disease stage (acute or chronic), the training level of the inspector and the prevalence of the disease in this area (i.e. lesion more or less known by inspectors).

Finally, as demonstrated before, a huge amount of data are available from slaughterhouses with a lot of different lesions, condemnation portions and zootechnical information that make definition of relevant groups of lesions difficult. The statistical approach presented in this study is one of the answers to this issue.

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